

Q1: 24 Feb (Shift 1) - Single Correct

The area (in sq. units) of the part of the circle $x^2 + y^2 = 36$, which is outside the parabola $y^2 = 9x$, is :

- (1) $24\pi + 3\sqrt{3}$
- (2) $12\pi + 3\sqrt{3}$
- (3) $12\pi - 3\sqrt{3}$
- (4) $24\pi - 3\sqrt{3}$

Q2: 24 Feb (Shift 2) - Single Correct

The area of the region : $R = \{(x, y) : 5x^2 \leq y \leq 2x^2 + 9\}$ is:

- (1) $9\sqrt{3}$ square units
- (2) $12\sqrt{3}$ square units
- (3) $11\sqrt{3}$ square units
- (4) $6\sqrt{3}$ square units

Q3: 25 Feb (Shift 1) - Numerical

The graphs of sine and cosine functions, intersect each other at a number of points and between two consecutive points of intersection, the two graphs enclose the same area A. Then A^4 is equal to

Q4: 26 Feb (Shift 1) - Numerical

The area bounded by the lines $y = ||x - 1| - 2|$ is

Note: NTA has dropped this question in the final official answer key.

Q5: 26 Feb (Shift 2) - Single Correct

Let A_1 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$ and y -axis in the first quadrant.

Also, let A_2 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, x -axis and $x = \frac{\pi}{2}$ in the first quadrant. Then,

- (1) $A_1 = A_2$ and $A_1 + A_2 = \sqrt{2}$

Questions with Answer Keys

MathonGo

(2) $A_1 : A_2 = 1 : 2$ and $A_1 + A_2 = 1$

(3) $2 A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$

(4) $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$

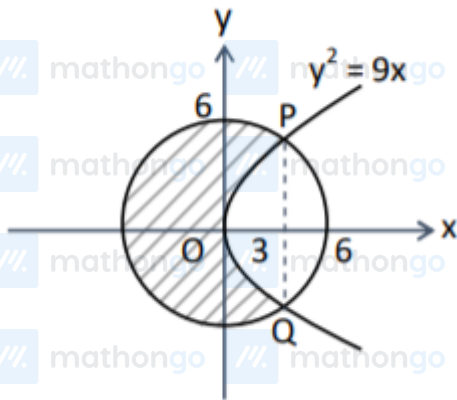
Answer Key**Q1 (4)****Q2 (2)****Q3 (64)****Q4 (8)****Q5 (4)**

Q1 (4)

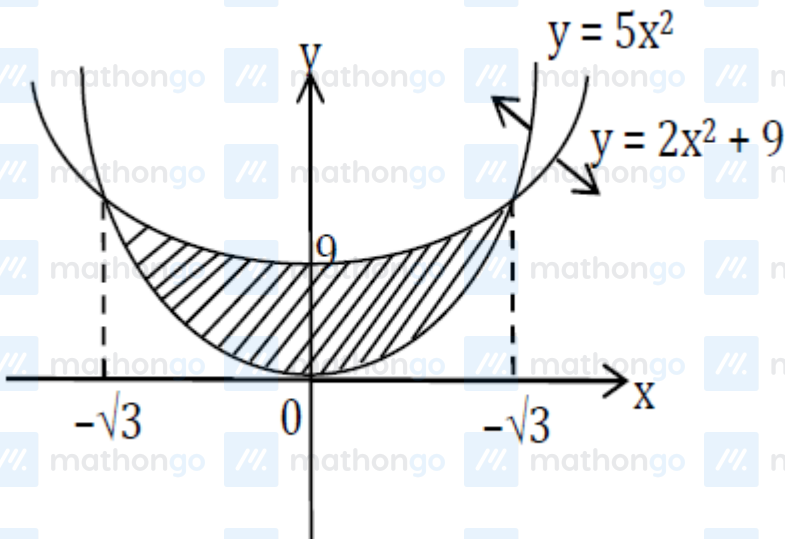
The curves intersect at point $(3, \pm 3\sqrt{3})$

Required area

$$\begin{aligned}
 &= \pi r^2 - 2 \left[\int_0^3 \sqrt{9x} dx + \int_3^6 \sqrt{36 - x^2} dx \right] \\
 &= 36\pi - 12\sqrt{3} - 2 \left(\frac{x}{2} \sqrt{36 - x^2} + 18 \sin^{-1} \left(\frac{x}{6} \right) \right)_3^6 \\
 &= 36\pi - 12\sqrt{3} - 2 \left(9 - \left(\frac{9\sqrt{3}}{2} + 3\pi \right) \right) = 24\pi - 3\sqrt{3}
 \end{aligned}$$



Q2 (2)



Required area

$$= 2 \int_0^{\sqrt{3}} (2x^2 + 9 - 5x^2) dx$$

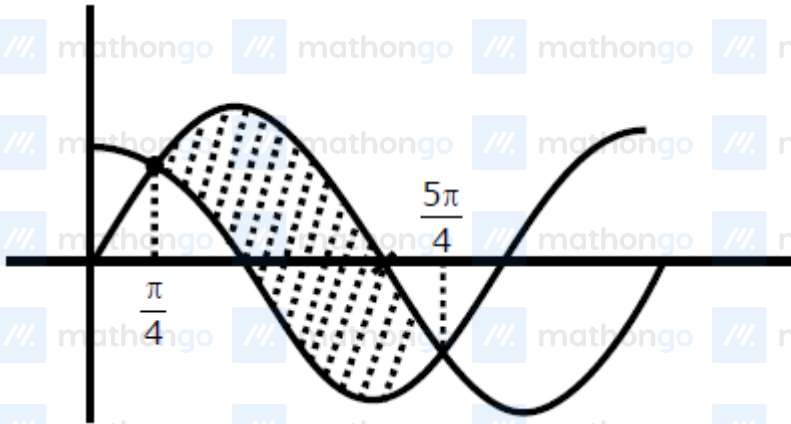
Hints and Solutions

MathonGo

$$= 2 \int_0^{\sqrt{3}} (9 - 3x^2) dx$$

$$= 2 \left[9x - x^3 \right]_0^{\sqrt{3}} = 12\sqrt{3}$$

Q3 (64)



$$A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx = [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$$

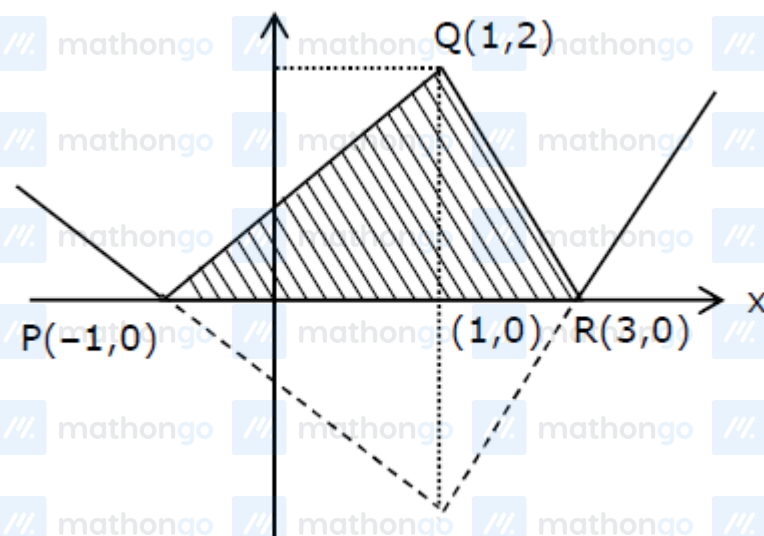
$$= - \left[\left(\cos \frac{5\pi}{4} + \sin \frac{\pi}{4} \right) - \left(\cos \frac{\pi}{4} + \sin \frac{\pi}{4} \right) \right]$$

$$= - \left[\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \right]$$

$$= \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

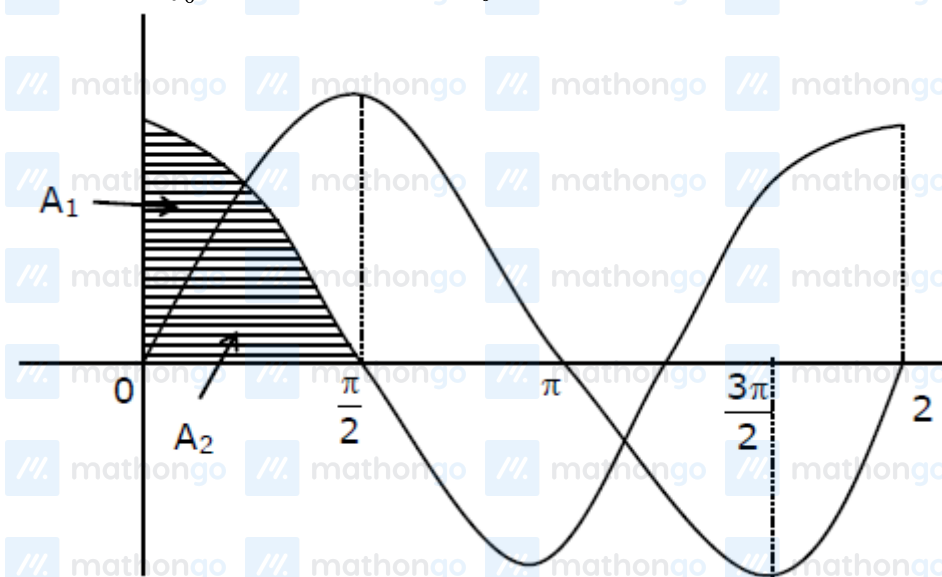
$$\Rightarrow A^4 = (2\sqrt{2})^4 = 64$$

Q4 (8)



Q5 (4)

$$A_1 + A_2 = \int_0^{\pi/2} \cos x \cdot dx = \sin x \Big|_0^{\pi/2} = 1$$



$$A_1 = \int_0^{\pi/4} (\cos x - \sin x) dx = (\sin x + \cos x) \Big|_0^{\pi/4} = \sqrt{2} - 1$$

$$\therefore A_2 = 1 - (\sqrt{2} - 1) = 2 - \sqrt{2}$$

$$\therefore \frac{A_1}{A_2} = \frac{\sqrt{2}-1}{\sqrt{2}(\sqrt{2}-1)} = \frac{1}{\sqrt{2}}$$